

Article

Optimization Model-Based Reliability Assessment for Distribution System with Data Centers Considering Multiple Heterogeneous Faults

Weiliang Zhong ¹, Yongbiao Liang ¹, Wei Huang ¹, Lieliang Hu ¹, Kaining Pan ¹, Sineng Li ¹, Qingjian Li ², Tao Yu ² and Wencong Xiao ^{2,*}

¹ Shaoguan Power Supply Bureau of Guangdong Power Grid Co., Ltd., Shaoguan 512000, China; zwl723964241@163.com (W.Z.); lywatch@163.com (Y.L.); hw498981675@163.com (W.H.); hll275923149@163.com (L.H.); pkn290417223@163.com (K.P.); lsn_549142455@163.com (S.L.)

² School of Electric Power Engineering, South China University of Technology, Guangzhou 510641, China; lqj_working@163.com (Q.L.); taoyu1@scut.edu.cn (T.Y.)

* Correspondence: 202311083113@mail.scut.edu.cn

Abstract

The integration of data centers (DCs) into distribution systems introduces new challenges for reliability assessment. It also provides operational flexibility through the spatial migration of computational workloads. However, existing reliability assessment methods have not sufficiently captured the joint effects of DC workload flexibility, post-fault network reconfiguration, and multiple heterogeneous faults. To address this gap, this paper proposes an optimization model-based reliability assessment framework for distribution systems with integrated DCs. A DC workload dispatch model is first developed by considering server operation, cooling demand, backup requirements, and inter-DC workload transfer limits. The post-fault operation process is then formulated as a two-stage optimization problem, including fault isolation and service restoration. The proposed framework co-optimizes network reconfiguration and DC workload migration under line faults, node faults, and remote-controlled switch failures. The resulting problem is reformulated as a mixed-integer linear programming model and solved using commercial solvers. Case studies on a modified IEEE 123-node test feeder show that DC workload migration can significantly reduce expected energy not supplied by exploiting the spatial flexibility of interconnected DCs. The results also demonstrate that neglecting node faults and switch failures may lead to overly optimistic reliability estimates. Additional tests on a modified 356-node regional distribution system further verify the applicability and scalability of the proposed framework. The proposed method provides a physically interpretable reliability assessment tool for active distribution systems with DC integration and heterogeneous fault mechanisms.



Academic Editor: Daniel Morinigo-Sotelo

Received: 29 April 2026

Revised: 25 June 2026

Accepted: 8 July 2026

Published: 13 July 2026

Copyright: © 2026 by the authors.

Licensee MDPI, Basel, Switzerland.

This article is an open access article distributed under the terms and conditions of the [Creative Commons Attribution \(CC BY\)](https://creativecommons.org/licenses/by/4.0/) license.

Keywords: data center workload migration; distribution system reliability assessment; heterogeneous faults; network reconfiguration; service restoration

1. Introduction

The rapid proliferation of the Internet of Things, artificial intelligence, and cloud computing has accelerated the expansion of data centers (DCs) as the primary computational backbone [1,2]. However, the continuous operation and intensive energy consumption of DCs pose significant challenges to distribution networks. In 2021, the total electricity

consumption of DCs in China reached 216.6 TWh, representing approximately 2.99% of the national total demand [3,4]. This figure is projected to exceed 700 TWh by 2035. As critical infrastructure for digital transformation, DCs support essential sectors such as healthcare and transportation [5,6]. Consequently, power supply interruptions triggered by distribution network contingencies directly jeopardize the operational security of DCs. Such outages entail substantial socio-economic losses and potentially catastrophic systemic risks. Therefore, it is imperative to develop precise reliability assessment models for distribution systems that explicitly account for DC integration.

DCs are characterized by intensive power demand, concentrated spatial distribution, and highly controllable operational attributes. Crucially, DCs possess significant energy flexibility [7]. Leveraging cloud interconnections and data center networks, interconnected DCs enable the spatial migration of computational workloads. This capability effectively reconfigures the power load distribution across the distribution network. In the event of network contingencies that threaten DC operations, computational tasks can be transferred to unaffected sites [8,9]. Such proactive load redistribution enhances overall system reliability and ensures the secure operation of the power grid.

Distribution system reliability assessment has been extensively studied. Existing methods can be broadly classified into analytical methods, Monte Carlo simulations, and optimization-based approaches. Analytical methods are computationally efficient and can provide closed-form or semi-analytical reliability indices [10–12]. They are suitable for systems with relatively fixed topology, predefined restoration rules, and limited uncertainty representation. Monte Carlo simulation is more flexible in modeling stochastic component failures, repair processes, renewable generation uncertainty, and time-varying operating conditions [13–15]. It should be noted that modern Monte Carlo-based reliability assessment is not limited to purely heuristic restoration logic. Some recent implementations incorporate optimization-based restoration models to better represent post-fault operational decisions [16]. Nevertheless, analytical and simulation-based methods may still face limitations when complex restoration actions, multiple flexible resources, and cyber-physical coupling need to be represented explicitly.

To address these limitations, optimization-based reliability assessment has attracted increasing attention in recent years [17–21]. These methods embed post-fault restoration decisions into explicit mathematical programming models. For example, topology-variable-based linear formulations have been developed to evaluate distribution system reliability while considering switching interruptions [17]. MILP-based models have also been proposed for reliability-constrained distribution network planning and expansion, allowing reliability indices to be directly incorporated into planning decisions [18,19]. More recent studies have further extended this line of research. Li et al. developed a linear-programming-based reliability assessment method that explicitly considers distribution automation and distributed generation [22]. Zhang et al. proposed a generalized MILP model for reliability evaluation of cabled active distribution networks with multiple devices [23]. In addition, cyber-physical factors and recovery resource decisions have been incorporated into recent reliability assessment frameworks for cyber-physical distribution systems [24]. These optimization-based approaches can explicitly represent network reconfiguration, load shedding, and service restoration decisions. Therefore, they provide a rigorous framework for quantifying the reliability contribution of post-fault operational flexibility.

Despite these advances, existing optimization-based reliability studies still have several limitations when applied to active distribution systems with integrated DCs. First, most existing models focus on physical-network-side resources, such as switches, distribution automation devices, distributed generators, and energy storage systems. These models can explicitly represent network reconfiguration, load shedding, and service restoration.

However, they do not fully capture the operational flexibility provided by interconnected DCs. In particular, the spatial migration of computational workloads can change the effective electrical demand at different locations after faults. This cyber-layer flexibility is different from conventional load shedding or local service restoration.

Second, recent studies on cyber-physical reliability and machine-learning-assisted reliability assessment have highlighted the importance of communication systems, data-driven prediction, and intelligent restoration decisions [25,26]. These studies provide useful tools for modeling cyber-physical interactions and improving computational efficiency. Nevertheless, the coordination between cyber-layer computational resource scheduling and physical-layer distribution network restoration remains insufficiently represented in explicit reliability optimization models. As a result, the reliability contribution of DC workload migration cannot be directly quantified by existing frameworks.

Third, many distribution system reliability studies still represent contingencies mainly through line outages or simplified component-failure assumptions. Such simplifications are suitable for conventional reliability studies, but they may be insufficient for modern active distribution systems with extensive automation devices. In practice, node faults and remote-controlled switch failures may change the feasible isolation actions, reduce the available restoration paths, and enlarge the outage area. Ignoring these heterogeneous fault mechanisms may therefore lead to overly optimistic reliability estimates.

To address the above research gaps, this paper proposes an optimization model-based reliability assessment framework for distribution systems with integrated DCs. The proposed framework is not intended to replace existing analytical, Monte Carlo-based, or optimization-based reliability assessment methods. Instead, it extends the optimization-based reliability assessment paradigm to a new cyber-physical operating context in which DC workload migration interacts with post-fault network restoration. The key distinction of this work lies in the unified modeling of three coupled factors: physical-layer network reconfiguration, cyber-layer DC workload migration, and multiple heterogeneous fault mechanisms. Under the assumptions of linearized distribution power flow, equivalent DC workload-transfer capacity, and MILP-representable post-fault operation constraints, the proposed model evaluates reliability by explicitly optimizing the fault isolation and service restoration processes. This enables the reliability contribution of DC spatial flexibility and the risk impact of node and switch failures to be quantified within the same mathematical framework.

The main contributions of this paper are summarized as follows:

1. An optimization model-based reliability assessment framework is developed for active distribution systems with integrated DCs. Different from conventional reliability assessment models that mainly evaluate physical load restoration, the proposed framework incorporates DC workload migration into post-fault reliability evaluation. This allows the cyber-layer flexibility of interconnected DCs to be quantified in terms of its impact on expected energy not supplied.
2. Multiple heterogeneous faults are explicitly incorporated into the reliability assessment framework, including line faults, node faults, and remote-controlled switch failures. This extends conventional line-fault-oriented reliability evaluation and enables a more realistic assessment of how different component failures affect fault isolation, service restoration, network topology, and reliability indices.
3. The proposed reliability assessment problem is reformulated as an MILP model and tested on modified IEEE 123-node and 356-node distribution systems. The case studies quantify the marginal contribution of DC workload migration, the assessment bias caused by neglecting heterogeneous faults, and the scalability of the proposed formulation. These results provide practical insights for reliability-oriented planning and post-fault operation of active distribution systems with DC integration.

The remainder of this paper is organized as follows: Section 2 establishes a high-fidelity model for DC workload dispatch. Section 3 develops an optimization-based reliability assessment model for distribution systems integrated with DCs, where multiple heterogeneous faults are explicitly incorporated. Numerical results and case studies are presented in Section 4 to validate the effectiveness of the proposed model. Finally, concluding remarks and future work are summarized in Section 5.

2. Modeling of Data Center Workload Dispatch

Geographically dispersed DCs can be interconnected to enable coordinated workload allocation and migration. Such spatial flexibility of DC loads can be leveraged to enhance the reliability of distribution networks. As illustrated in Figure 1, the computational workload of DC 1 can be transferred to DC 2 and DC 3 during a power outage at DC 1. This spatial task migration effectively reduces the unserved energy and facilitates the post-fault restoration of the distribution system.

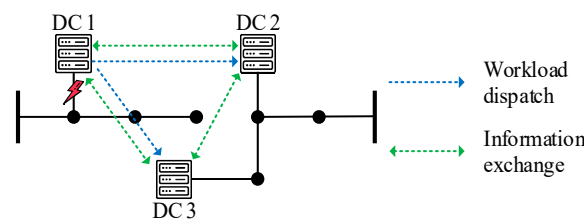


Figure 1. Workload dispatch for post-fault restoration in distribution networks.

However, the inter-DC workload migration is not unconstrained. The spatial task transfer is strictly governed by multi-dimensional factors, including data volume, computational resource availability, and communication bandwidth capacities. Consequently, these constraints must be explicitly incorporated into the reliability assessment model to ensure feasible and realistic restoration strategies.

The primary energy-consuming components of a DC include servers, cooling systems, and lighting facilities. Among these, servers and cooling infrastructures collectively account for approximately 90% of the total electricity consumption. Consequently, this study focuses on the energy modeling of servers and cooling systems to characterize the dominant power demand of DCs while maintaining computational efficiency. The total power consumption of the i -th DC at time t is formulated as:

$$P_{i,s,t}^{\text{DC}} = (P_{i,s,t}^{\text{IT}} + P_{i,s,t}^{\text{cool}}) / \lambda^{\text{DC}} \quad (1)$$

$$0 \leq P_{i,s,t}^{\text{DC}} \leq P_i^{\text{DC,max}} \quad (2)$$

where $P_{i,s,t}^{\text{DC}}$, $P_{i,s,t}^{\text{IT}}$, $P_{i,s,t}^{\text{cool}}$ denote the total power demand, server power consumption, and cooling power of the DC at node i in scenario s at time step t , respectively. λ^{DC} is the power usage effectiveness coefficient of the DC. $P_i^{\text{DC,max}}$ signifies the maximum power capacity limit of the DC at node i .

The power consumption of the IT infrastructure in the i -th DC, denoted as $P_{i,s,t}^{\text{IT}}$, is formulated in (3). Specifically, it consists of the total static power of active servers and the incremental power proportional to the computational workload. The operational constraints of the servers are defined in (4) and (5):

$$P_{i,s,t}^{\text{IT}} = n_{i,s,t}^{\text{IT-on}} P_i^{\text{IT-i}} + (P_i^{\text{IT-P}} - P_i^{\text{IT-i}}) \frac{W_{i,s,t}}{\varphi_i} \quad (3)$$

$$0 \leq n_{i,s,t}^{\text{IT-on}} \leq (1 - \phi) n_i^{\text{IT}} \quad (4)$$

$$0 \leq \frac{W_{i,s,t}}{n_{i,s,t}^{\text{IT-on}} \varphi_i} \leq S_i^{\max} \quad (5)$$

where $P_i^{\text{IT-i}}$ and $P_i^{\text{IT-P}}$ represent the idle and peak power consumption of a single server, respectively; $n_{i,s,t}^{\text{IT-on}}$ is the integer decision variable representing the number of active servers; $W_{i,s,t}$ denotes the computational workload assigned to the DC; n_i^{IT} and ϕ signify the total number of installed servers and the backup coefficient, respectively; $S_i^{\max}$ is the upper bound of the CPU utilization for a single server; φ_i represents the processing capacity or service rate of each server unit.

Specifically, (4) enforces the capacity limit on active servers, where a certain proportion of the total installed servers, denoted by the backup coefficient ϕ , is reserved for redundancy and maintenance purposes to ensure high availability. (4) represents the CPU utilization constraint, which ensures that the average computational intensity of active servers remains within a safe and efficient operational range. By imposing the upper bound $S_i^{\max}$, the model prevents server performance degradation and hardware overheating caused by excessive computational stress.

The power demand of the cooling system and its associated capacity constraints are formulated as follows:

$$P_{i,s,t}^{\text{cool}} = H_{i,s,t}^{\text{cool}} / \lambda^{\text{cool}} \quad (6)$$

$$P_{i,s,t}^{\text{cool}} \leq P_i^{\text{cool,max}} \quad (7)$$

where $H_{i,s,t}^{\text{cool}}$ represents the heat load generated by the IT equipment at node i in scenario s at time t ; λ^{cool} denotes the coefficient of performance of the cooling system. $P_i^{\text{cool,max}}$ signifies the rated power capacity of the installed cooling infrastructure at node i .

Next, the workload dispatch process of data centers is modeled. Interconnected DCs enable geographical workload migration. Under fault conditions, part of the workload of an outage-affected DC can be transferred to other unaffected DCs. This process improves service continuity and reduces the impact of power interruptions. The workload dispatch is expressed as:

$$P_{i,s,t+1}^{\text{DC}} = P_{i,s,t}^{\text{DC}} - \sum_{j \neq i} P_{ij,s,t}^{\text{DC}} + \sum_{j \neq i} P_{ji,s,t}^{\text{DC}} \quad (8)$$

where $P_{ij,s,t}^{\text{DC}}$ denotes the workload transferred from DC i to DC j .

To ensure the feasibility of inter-DC workload migration, the amount of migrated workload should not exceed the transfer capability supported by the communication link between DC i and DC j . In this study, $P_{ij,s,t}^{\text{DC}}$ denotes the migrated workload from DC i to DC j , expressed as the equivalent electrical power demand of the corresponding computational workload. Accordingly, B_{ij}^{DC} is defined as the maximum transferable equivalent DC load allowed by the inter-DC communication link. It is not the raw communication bandwidth in bit/s, but an equivalent workload-transfer capacity after converting the available communication bandwidth, latency requirement, and workload-to-power relationship into the same unit as $P_{ij,s,t}^{\text{DC}}$. Therefore, the workload migration constraint is formulated as

$$P_{ij,s,t}^{\text{DC}} \leq B_{ij}^{\text{DC}} \quad (9)$$

Here, B_{ij}^{DC} has the same unit as $P_{ij,s,t}^{\text{DC}}$, for example MW, and represents the upper bound of the equivalent DC load that can be transferred from DC i to DC j within the considered time interval. When the constraint in (9) becomes binding, the amount of workload that can be migrated from the affected DC to remote DCs is limited by the available equivalent transfer capability. The remaining DC demand must then be supplied locally through

network reconfiguration if feasible; otherwise, it is treated as unsupplied demand in the reliability assessment.

In summary, (1)–(7) constitute the proposed DC workload dispatch model. This model captures the interconnection among data centers and enables the geographical migration of computational workloads. It provides the modeling foundation for incorporating DC flexibility into the subsequent distribution system reliability assessment.

3. Optimization Model-Based Reliability Assessment

In this section, an optimization model-based reliability assessment method is developed for active distribution systems with multiple resources. The model explicitly considers the integration of DCs and distributed generators (DGs). In particular, it incorporates the DC workload dispatch model proposed in Section 2 to capture workload migration under fault conditions. In addition, the proposed method accounts for multiple heterogeneous faults, including failures of distribution lines, substations, and switches.

To achieve a more accurate reliability assessment, the post-fault operation process of the distribution system is divided into two stages: fault isolation (FI) and service restoration (SR). Figure 2 illustrates a typical post-fault operation process. During normal operation (NO), a fault occurs on branch 3–4. In the FI stage, branches 3–4 and 4–5 are disconnected to isolate the fault. As a result, an island consisting of DG 8, node 4, and DC 9 is formed. Owing to the limited capacity of DG 8, it cannot simultaneously supply DC 9 and the loads at nodes 4, 5, and 11. Therefore, the loads at nodes 5 and 11 are interrupted, and part of the workload of DC 9 is migrated to DC 10. In the SR stage, the switches on branches 4–5 and 5–6 are closed. Through network reconfiguration, power supply to all loads is successfully restored.

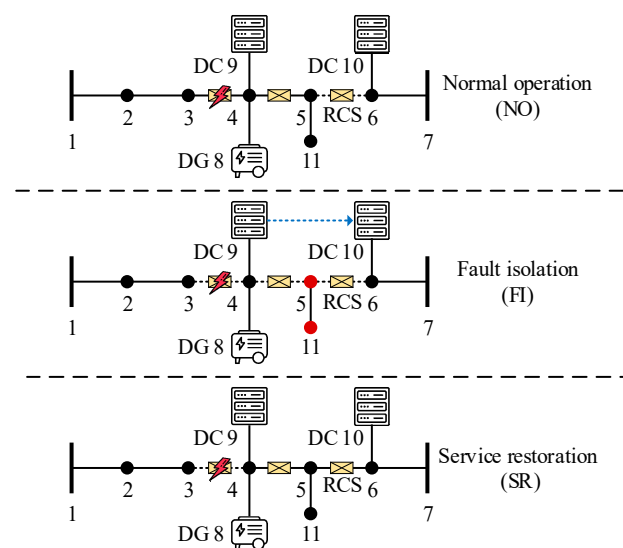


Figure 2. Post-fault operation process of the distribution system.

3.1. Objective Function

The expected energy not supplied (EENS) is adopted as the reliability index in the proposed assessment model. It quantifies the expected amount of unsupplied energy caused by system faults and consequent outages over the evaluation horizon. Accordingly, the objective function is formulated as:

$$\min \text{EENS} = \sum_{s \in S} p_s \sum_{PH \in \{FI, SR\}} \text{ENS}_s^{PH} \quad (10)$$

where p_s denotes the occurrence probability of scenario s ; ENS_s^{PH} denotes the unsupplied energy in scenario s at stage PH , including FI and SR; S represents the sets of all scenarios.

The unsupplied energy ENS_s^{PH} in scenario s at stage PH is further calculated as:

$$ENS_s^{PH} = \sum_{i \in \mathcal{N}} P_{i,s}^{L,PH} \cdot (1 - y_{i,s}^{PH}) \cdot \Delta t \quad (11)$$

where $P_{i,s,t}^{L,PH}$ denotes the load demand at node i in scenario s at stage PH , and $y_{i,s}^{PH}$ is a binary variable indicating the energization status of node i . Specifically, $y_{i,s}^{PH} = 1$ indicates that node i is energized, while $y_{i,s}^{PH} = 0$ indicates that node i is de-energized.

In the proposed formulation, the trade-off between local electrical load restoration and remote DC workload migration is determined endogenously by the optimization model. The objective function minimizes EENS, while the constraints jointly describe network reconfiguration, nodal energization, DG output, DC workload balance, and inter-DC workload transfer capability. Therefore, the model does not impose a predefined priority between restoring loads locally and migrating computational workloads remotely. Instead, it selects the feasible combination of switching actions, nodal energization decisions, and workload transfer decisions that minimizes the total unsupplied energy under both physical-network and cyber-layer constraints.

3.2. Fault Isolation Constraints

Conventional distribution system reliability assessment usually considers only branch faults and neglects the presence of multiple heterogeneous fault types, which may lead to inaccurate evaluation results. In contrast, the proposed method explicitly considers multiple types of faults. In addition to branch faults, node faults, and remote-controlled switch (RCS) failures are also incorporated. A fault indicator variable is introduced as follows:

$$\mathcal{F} = \{F_{ij,s}^B, F_{i,s}^N, F_{ij,s}^{RCS}\} \quad (12)$$

where $F_{ij,s}^B$, $F_{i,s}^N$, $F_{ij,s}^{RCS}$ denote the fault states of branch ij , node i , and RCS on branch ij in scenario s , respectively. A value of 1 indicates that the corresponding component is faulty, while a value of 0 indicates normal operation. This definition applies to all considered equipment types, including lines, nodes, and switches. Once the set of critical fault scenarios is determined, F is treated as a parameter in the proposed model.

(1) Power flow constraints: The linearized DistFlow model is adopted in this paper [27]. Equations (13) and (14) describe the active and reactive power balance, while (15) represents the linearized voltage drop constraint [28]. Equations (16)–(19) indicate that load at de-energized nodes will be shed.

$$\sum_{k:(i,k)} P_{ik,s}^{FI} - \sum_{k:(k,i)} P_{ki,s}^{FI} = P_{i,s}^{S,FI} + P_{i,s}^{DG,FI} - P_{i,s}^{DC,FI} - P_{i,s}^{L,FI} \quad (13)$$

$$\sum_{k:(i,k)} Q_{ik,s}^{FI} - \sum_{k:(k,i)} Q_{ki,s}^{FI} = Q_{i,s}^{S,FI} + Q_{i,s}^{DG,FI} - Q_{i,s}^{DC,FI} - Q_{i,s}^{L,FI} \quad (14)$$

$$-M(1 - z_{ij,s}^{FI}) \leq U_{i,s}^{FI} - U_{j,s}^{FI} - 2(P_{ij,s}^{FI} r_{ij} + Q_{ij,s}^{FI} x_{ij}) \leq M(1 - z_{ij,s}^{FI}) \quad (15)$$

$$P_{i,s}^{L,FI} \leq P_{i,s}^{L,FI} y_{i,s}^{FI} \quad (16)$$

$$Q_{i,s}^{L,FI} \leq Q_{i,s}^{L,FI} y_{i,s}^{FI} \quad (17)$$

$$P_{i,s}^{DC,FI} \leq P_{i,s}^{DC,FI} y_{i,s}^{FI} \quad (18)$$

$$Q_{i,s}^{DC,FI} \leq Q_{i,s}^{DC,FI} y_{i,s}^{FI} \quad (19)$$

where $P_{ik,s}^{\text{FI}}$ and $Q_{ik,s}^{\text{FI}}$ denote the active and reactive power flow on branch ik during the FI phase of scenario s ; $P_{i,s}^{\text{S,FI}}$ and $Q_{i,s}^{\text{S,FI}}$ denote the active and reactive power injection from the substation at node i during the FI phase of scenario s ; $P_{i,s}^{\text{DG,FI}}$ and $Q_{i,s}^{\text{DG,FI}}$ denote the active and reactive power injection from the DG at node i during the FI phase of scenario s ; $P_{i,s}^{\text{L,FI}}$ and $Q_{i,s}^{\text{L,FI}}$ denote the active and reactive power demand at node i during the FI phase of scenario s ; $P_{i,s}^{\text{DC,FI}}$ and $Q_{i,s}^{\text{DC,FI}}$ denote the active and reactive power demand of the DC at node i during the FI phase of scenario s ; r_{ij} and x_{ij} denote the resistance and inductance of branch ij ; M denotes a large number; $U_{i,s}^{\text{FI}}$ denotes the square of voltage magnitude at node i during the FI phase of scenario s ; $z_{ij,s}^{\text{FI}}$ is a binary variable, which denotes the status of branch ij during the FI phase of scenario s . It should be noted that a DC is essentially a type of load. However, it is modeled separately here to facilitate the representation of workload dispatch.

(2) Secure operation constraints: Equations (20) and (21) define the power output constraints of DGs. The nonlinear branch power flow model is linearized in (22)–(25) [28]. Equation (26) imposes the nodal voltage constraint.

$$P_i^{\text{DG,min}} y_{i,s}^{\text{FI}} \leq P_{i,s}^{\text{DG,FI}} \leq P_i^{\text{DG,max}} y_{i,s}^{\text{FI}} \quad (20)$$

$$Q_i^{\text{DG,min}} y_{i,s}^{\text{FI}} \leq Q_{i,s}^{\text{DG,FI}} \leq Q_i^{\text{DG,max}} y_{i,s}^{\text{FI}} \quad (21)$$

$$-S_{ij}^{\text{max}} z_{ij,s}^{\text{FI}} \leq P_{ij,s}^{\text{FI}} \leq S_{ij}^{\text{max}} z_{ij,s}^{\text{FI}} \quad (22)$$

$$-S_{ij}^{\text{max}} z_{ij,s}^{\text{FI}} \leq Q_{ij,s}^{\text{FI}} \leq S_{ij}^{\text{max}} z_{ij,s}^{\text{FI}} \quad (23)$$

$$-\sqrt{2} S_{ij}^{\text{max}} z_{ij,s}^{\text{FI}} \leq P_{ij,s}^{\text{FI}} + Q_{ij,s}^{\text{FI}} \leq \sqrt{2} S_{ij}^{\text{max}} z_{ij,s}^{\text{FI}} \quad (24)$$

$$-\sqrt{2} S_{ij}^{\text{max}} z_{ij,s}^{\text{FI}} \leq P_{ij,s}^{\text{FI}} - Q_{ij,s}^{\text{FI}} \leq \sqrt{2} S_{ij}^{\text{max}} z_{ij,s}^{\text{FI}} \quad (25)$$

$$|U_i^{\text{min}}|^2 \leq U_{i,s}^{\text{FI}} \leq |U_i^{\text{max}}|^2 \quad (26)$$

where $P_i^{\text{DG,max}}$ and $P_i^{\text{DG,min}}$ denote the maximum and minimum active power outputs of the DG at node i , respectively; $Q_i^{\text{DG,max}}$ and $Q_i^{\text{DG,min}}$ denote the maximum and minimum reactive power outputs of the DG at node i , respectively; S_{ij}^{max} denotes the maximum capacity of branch ij ; U_i^{max} and U_i^{min} denote the upper and lower bounds of the voltage at node i .

(3) Radial topology constraints: Distribution systems must maintain a radial topology. To reduce computational complexity, the radiality constraints are formulated using continuous variables [29]. Owing to the controllability of DGs, intentional islanded operation is allowed [26]. In addition, since coordinated operation between DGs and substations is supported, nodes with DGs are treated as regular load nodes in the radiality constraints.

$$f_{ij,s}^{+, \text{FA}} + f_{ij,s}^{-, \text{FA}} = z_{ij,s}^{\text{FA}} \quad (27)$$

$$\sum_{j:j \rightarrow i} f_{ji,s}^{+, \text{FA}} + \sum_{j:i \rightarrow j} f_{ij,s}^{+, \text{FA}} \leq \frac{|\Omega_N| - |\Omega_S|}{|\Omega_N| - |\Omega_S| + 1}, \forall i \notin \Omega_S \quad (28)$$

$$\sum_{i \in \Omega_S} \left[\sum_{j:j \rightarrow i} f_{ji,s}^{+, \text{FA}} + \sum_{j:i \rightarrow j} f_{ij,s}^{+, \text{FA}} \right] \leq \frac{|\Omega_N| - |\Omega_S|}{|\Omega_N| - |\Omega_S| + 1} \quad (29)$$

where $f_{ij,s}^{+, \text{FI}}$ and $f_{ij,s}^{-, \text{FI}}$ denote the auxiliary variables for the radial constraint on branch ij during the FI phase of scenario s ; Ω_N and Ω_S denote the set of all nodes and the set of substation nodes, respectively; $|\cdot|$ denotes the size of the set.

(4) Switching operation constraints

$$1 - x_{ij}^{\text{RCS}} \leq z_{ij,s}^{\text{FI}} \quad (30)$$

$$z_{ij,s}^{\text{FI}} \leq z_{ij}^{\text{NO}} \quad (31)$$

$$z_{ij,s}^{\text{FI}} \leq 2 - F_{ij,s}^{\text{B}} - x_{ij}^{\text{RCS}}(1 - F_{ij,s}^{\text{RCS}}) \quad (32)$$

$$z_{ij,s}^{\text{FI}} = z_{ij}^{\text{NO}} F_{ij,s}^{\text{RCS}} \quad (33)$$

$$n_{i,s}^{\text{FI}} \geq F_{ij,s}^{\text{B}} + z_{ij,s}^{\text{FI}} - 1 \quad (34)$$

$$n_{j,s}^{\text{FI}} \geq F_{ij,s}^{\text{B}} + z_{ij,s}^{\text{FI}} - 1 \quad (35)$$

$$n_{i,s}^{\text{FI}} = F_{i,s}^{\text{N}} \quad (36)$$

$$n_{i,s}^{\text{FI}} + z_{ij,s}^{\text{FI}} - 1 \leq n_{j,s}^{\text{FI}} \quad (37)$$

$$n_{j,s}^{\text{FI}} + z_{ij,s}^{\text{FI}} - 1 \leq n_{i,s}^{\text{FI}} \quad (38)$$

$$y_{i,s}^{\text{FI}} + z_{ij,s}^{\text{FI}} - 1 \leq y_{j,s}^{\text{FI}} \quad (39)$$

$$y_{j,s}^{\text{FI}} + z_{ij,s}^{\text{FI}} - 1 \leq y_{i,s}^{\text{FI}} \quad (40)$$

$$y_{j,s}^{\text{FI}} \leq 1 - n_{j,s}^{\text{FI}} \quad (41)$$

$$y_{j,s}^{\text{FI}} = 1, j \in \Omega_S \quad (42)$$

where x_{ij}^{RCS} is a binary variable indicating whether an RCS is installed on branch ij . A value of $x_{ij}^{\text{RCS}} = 1$ indicates that an RCS is installed, while $x_{ij}^{\text{RCS}} = 0$ indicates that no RCS is installed. z_{ij}^{NO} denotes the status of branch ij at the normal operation phase; $n_{i,s}^{\text{FI}}$ is a binary variable indicating whether a node i is involved in a faulted zone. A value of $n_{i,s}^{\text{FI}} = 1$ indicates that a node i is involved in a faulted zone, while $n_{i,s}^{\text{FI}} = 0$ indicates that the node is not involved in a faulted zone.

Equation (30) enforces that branches without RCSs remain closed. Equation (31) indicates that, during the FI stage, switches can only be opened and cannot be closed. Equation (32) ensures that if branch ij is faulty and equipped with an RCS, and the RCS is not faulty, the corresponding switch is opened to isolate the fault. Equation (33) indicates that if an RCS fails, its switching status in the FI stage remains the same as that in the NO stage. Equations (34) and (35) indicate that if branch ij is faulty and remains closed, both terminal nodes i and j are classified into the faulted zone. Equation (36) indicates that a node is classified as being in the faulted zone if it experiences a fault. Equations (37) and (38) ensure that the two terminal nodes of a closed branch belong to the same zone, namely, either the faulted zone or the non-faulted zone. Equations (39) and (40) enforce that the two terminal nodes of a closed branch have the same energization status. Equation (41) indicates that a node in the faulted zone must be de-energized. Equation (42) specifies that substation nodes are always energized.

In summary, during the FI stage, the proposed model isolates faults through remote switching of available RCSs, thereby reducing the outage area. Meanwhile, DCs can mitigate interrupted demand through workload dispatch. In addition, the proposed model considers branch, node, and switch failures, which enables a more refined reliability assessment.

3.3. Service Restoration Constraints

After fault isolation, fast service restoration is implemented through RCS-based switching to restore loads outside the faulted zones. The faulted zones must remain isolated from the non-faulted zones.

(1) Power flow constraints: Equations (43) and (44) represent the active and reactive power balance constraints, respectively. Equation (45) describes the branch power flow equation. Equations (46)–(49) ensure that the loads at de-energized nodes are fully shed.

$$\sum_{k:(i,k)} P_{ik,s}^{\text{SR}} - \sum_{k:(k,i)} P_{ki,s}^{\text{SR}} = P_{i,s}^{\text{S,SR}} + P_{i,s}^{\text{DG,SR}} - P_{i,s}^{\text{DC,SR}} - P_{i,s}^{\text{L,SR}} \quad (43)$$

$$\sum_{k:(i,k)} Q_{ik,s}^{\text{SR}} - \sum_{k:(k,i)} Q_{ki,s}^{\text{SR}} = Q_{i,s}^{\text{S,SR}} + Q_{i,s}^{\text{DG,SR}} - Q_{i,s}^{\text{DC,SR}} - Q_{i,s}^{\text{L,SR}} \quad (44)$$

$$-M(1 - z_{ij,s}^{\text{SR}}) \leq U_{i,s}^{\text{SR}} - U_{j,s}^{\text{SR}} - 2(P_{ij,s}^{\text{SR}} r_{ij} + Q_{ij,s}^{\text{SR}} x_{ij}) \leq M(1 - z_{ij,s}^{\text{SR}}) \quad (45)$$

$$P_{i,s}^{\text{L,SR}} \leq P_{i,s}^{\text{L}} y_{i,s}^{\text{SR}} \quad (46)$$

$$Q_{i,s}^{\text{L,SR}} \leq Q_{i,s}^{\text{L}} y_{i,s}^{\text{SR}} \quad (47)$$

$$P_{i,s}^{\text{DC,SR}} \leq P_{i,s}^{\text{DC}} y_{i,s}^{\text{SR}} \quad (48)$$

$$Q_{i,s}^{\text{DC,SR}} \leq Q_{i,s}^{\text{DC}} y_{i,s}^{\text{SR}} \quad (49)$$

(2) Secure operation constraints: Equations (50) and (51) define the power output limits of DGs. Equations (52)–(55) specify the operational constraints of the network. Equation (56) imposes the voltage magnitude limits.

$$P_i^{\text{DG,min}} y_{i,s}^{\text{SR}} \leq P_{i,s}^{\text{DG,SR}} \leq P_i^{\text{DG,max}} y_{i,s}^{\text{SR}} \quad (50)$$

$$Q_i^{\text{DG,min}} y_{i,s}^{\text{SR}} \leq Q_{i,s}^{\text{DG,SR}} \leq Q_i^{\text{DG,max}} y_{i,s}^{\text{SR}} \quad (51)$$

$$-S_{ij}^{\text{max}} z_{ij,s}^{\text{SR}} \leq P_{ij,s}^{\text{SR}} \leq S_{ij}^{\text{max}} z_{ij,s}^{\text{SR}} \quad (52)$$

$$-S_{ij}^{\text{max}} z_{ij,s}^{\text{SR}} \leq Q_{ij,s}^{\text{SR}} \leq S_{ij}^{\text{max}} z_{ij,s}^{\text{SR}} \quad (53)$$

$$-\sqrt{2} S_{ij}^{\text{max}} z_{ij,s}^{\text{SR}} \leq P_{ij,s}^{\text{SR}} + Q_{ij,s}^{\text{SR}} \leq \sqrt{2} S_{ij}^{\text{max}} z_{ij,s}^{\text{SR}} \quad (54)$$

$$-\sqrt{2} S_{ij}^{\text{max}} z_{ij,s}^{\text{SR}} \leq P_{ij,s}^{\text{SR}} - Q_{ij,s}^{\text{SR}} \leq \sqrt{2} S_{ij}^{\text{max}} z_{ij,s}^{\text{SR}} \quad (55)$$

$$|U_i^{\text{min}}|^2 \leq U_{i,s}^{\text{SR}} \leq |U_i^{\text{max}}|^2 \quad (56)$$

(3) Radial topology constraints: The radial topology constraints are formulated as follows:

$$f_{ij,s}^{+, \text{SR}} + f_{ij,s}^{-, \text{SR}} = z_{ij,s}^{\text{SR}} \quad (57)$$

$$\sum_{j:j \rightarrow i} f_{ji,s}^{+, \text{SR}} + \sum_{j:i \rightarrow j} f_{ij,s}^{+, \text{SR}} \leq \frac{|\Omega_N| - |\Omega_S|}{|\Omega_N| - |\Omega_S| + 1}, \forall i \notin \Omega_S \quad (58)$$

$$\sum_{i \in \Omega_S} \left[\sum_{j:j \rightarrow i} f_{ji,s}^{+, \text{SR}} + \sum_{j:i \rightarrow j} f_{ij,s}^{+, \text{SR}} \right] \leq \frac{|\Omega_N| - |\Omega_S|}{|\Omega_N| - |\Omega_S| + 1} \quad (59)$$

(4) Switching operation constraints

$$1 - x_{ij}^{\text{RCS}} \leq z_{ij,s}^{\text{SR}} \quad (60)$$

$$z_{ij,s}^{\text{SR}} \leq 2 - F_{ij,s}^{\text{B}} - x_{ij}^{\text{RCS}} (1 - F_{ij,s}^{\text{RCS}}) \quad (61)$$

$$z_{ij,s}^{SR} = z_{ij}^{NO} F_{ij,s}^{RCS} \quad (62)$$

$$n_{i,s}^{SR} \geq F_{ij,s}^B + z_{ij,s}^{SR} - 1 \quad (63)$$

$$n_{j,s}^{SR} \geq F_{ij,s}^B + z_{ij,s}^{SR} - 1 \quad (64)$$

$$n_{i,s}^{SR} = F_{i,s}^N \quad (65)$$

$$n_{i,s}^{SR} + z_{ij,s}^{SR} - 1 \leq n_{j,s}^{SR} \quad (66)$$

$$n_{j,s}^{SR} + z_{ij,s}^{SR} - 1 \leq n_{i,s}^{SR} \quad (67)$$

$$y_{i,s}^{SR} + z_{ij,s}^{SR} - 1 \leq y_{j,s}^{SR} \quad (68)$$

$$y_{j,s}^{SR} + z_{ij,s}^{SR} - 1 \leq y_{i,s}^{SR} \quad (69)$$

$$y_{j,s}^{SR} \leq 1 - n_{j,s}^{SR} \quad (70)$$

$$y_{j,s}^{SR} = 1, j \in \Omega_S \quad (71)$$

$$y_{j,s}^{SR} \geq y_{j,s}^{FA} \quad (72)$$

As shown in (43)–(72), the constraints in the SR stage are largely consistent with those in the FI stage. Most variables have the same physical meanings as their FI-stage counterparts, and the superscripts FI and SR are used only to distinguish the corresponding post-fault operation stages. Specifically, variables with the superscript FI denote the decisions or states during the fault isolation stage, whereas variables with the superscript SR denote the corresponding decisions or states during the service restoration stage.

The main differences between the FI and SR formulations are as follows. First, constraint (31) in the FI stage is not included in the SR stage because the SR stage permits switch closing for load restoration, whereas the FI stage only allows switch opening for fault isolation. Second, constraint (72) requires that loads energized in the FI stage remain supplied in the SR stage. This prevents service interruption to already restored loads and preserves restoration continuity. Moreover, constraint (72) establishes the coupling between the FI and SR stages by linking the load energization status in the SR stage to that in the FI stage. Therefore, any load restored during fault isolation must remain energized during service restoration.

3.4. Mixed-Integer Linear Programming Formulation

The proposed model can be reformulated as a mixed-integer linear programming (MILP) problem. Its general form is given as follows:

$$\min \mathbf{b}^T \mathbf{x} + \mathbf{d}^T \mathbf{y} \quad (73)$$

$$\text{s.t. } \mathbf{Ax} + \mathbf{Dy} \geq \mathbf{e} \quad (74)$$

$$\mathbf{Fx} + \mathbf{Gy} = \mathbf{f} \quad (75)$$

where $\mathbf{x}$ and $\mathbf{y}$ denote the vectors of continuous and binary decision variables, respectively. $\mathbf{b}$ and $\mathbf{d}$ are the corresponding coefficient vectors in the objective function. $\mathbf{A}$, $\mathbf{D}$, $\mathbf{F}$, and $\mathbf{G}$ are coefficient matrices, while $\mathbf{e}$ and $\mathbf{f}$ are constant vectors.

Accordingly, (2), (4), (5), (7), (9), (15)–(26), (28)–(32), (34) and (35), (37)–(41), (45)–(56), (58)–(61), (63) and (64), (66)–(70), and (72) correspond to the inequality constraint (74); (1), (3), (6), (8), (13) and (14), (27), (33), (36), (42), (43) and (44), (57), (62), (65), and (71) correspond to the equality constraint (75); and (10) corresponds to the objective function (73). The resulting MILP problem can be solved by branch-and-bound or branch-and-cut

algorithms implemented in commercial solvers such as CPLEX and Gurobi. Reformulating the proposed reliability assessment model as an MILP improves computational tractability and facilitates the use of off-the-shelf optimization solvers. It also provides a flexible framework that can be extended to incorporate additional operational measures and modeling details.

The computational complexity of the proposed MILP formulation mainly depends on the network size, the number of controllable switches, the number of DCs, and the number of fault scenarios. For each fault scenario, increasing the number of nodes and branches introduces more power-flow variables, voltage variables, nodal energization variables, and branch-status variables. Increasing the number of RCSs enlarges the feasible reconfiguration space and adds switching-related binary decisions. Increasing the number of DCs introduces additional workload allocation and inter-DC migration variables, especially when workload transfer among multiple DC pairs is considered. Therefore, the solution time generally increases with the physical network scale, the number of controllable devices, and the cyber-layer workload migration dimension. Although MILP problems have exponential worst-case complexity with respect to the number of binary variables, the proposed reliability assessment problem is naturally separable across fault scenarios. Once the fault state is specified, the post-fault restoration problem for each scenario can be solved independently. This feature enables scenario-wise parallel computation and helps maintain computational tractability for larger systems.

3.5. Post-Evaluation of Reliability Indices

After the optimization problem is solved, the reliability indices can be further evaluated based on the obtained results. In this paper, the adopted reliability indices include the system average interruption duration index (SAIDI), system average interruption frequency index (SAIFI), average service availability index (ASAI), and expected energy not supplied (EENS). The corresponding formulations are given in (76)–(79).

$$SAIDI = \frac{\sum_{i \in \mathcal{N}} N_i W_i}{\sum_{i \in \mathcal{N}} N_i} \quad (76)$$

$$SAIFI = \frac{\sum_{i \in \mathcal{N}} \lambda_i N_i}{\sum_{i \in \mathcal{N}} N_i} \quad (77)$$

$$ASAI = \frac{8760 \sum_{i \in \mathcal{N}} N_i W_i - \sum_{i \in \mathcal{N}} N_i W_i}{8760 \sum_{i \in \mathcal{N}} N_i} \quad (78)$$

$$EENS = \frac{\sum_{i \in \mathcal{N}} P_i W_i}{\sum_{i \in \mathcal{N}} N_i} \quad (79)$$

Here, N_i denotes the number of customers at load point i , W_i denotes the equivalent annual average outage duration at load point i , λ_i denotes the annual failure rate at load point i , and P_i denotes the average annual load at load point i . The equivalent annual average outage duration W_i is calculated as follows:

$$W_i = \sum_{s \in \mathcal{S}} (1 - y_{i,s}^{\text{FI}}) * \tau^{\text{FI}} + (1 - y_{i,s}^{\text{SR}}) * \tau^{\text{SR}} \quad (80)$$

where τ^{FI} denotes the outage duration associated with the fault isolation stage when the node remains de-energized after fault isolation, and τ^{SR} denotes the outage duration

associated with the service restoration stage when the node is still not restored after service restoration. These two terms together characterize the outage duration experienced by the load point under a fault scenario.

4. Case Studies

In this section, the proposed reliability assessment method is validated on a modified IEEE 123-node test feeder [30] and a modified 356-node regional system in China [31]. All simulations are conducted on a personal computer equipped with an Intel Core i5 quad-core processor at 3.00 GHz and 8 GB of RAM. The model is implemented in Python 3.11 and solved using Gurobi 11.0. The relative optimality gap is set to 0.01%.

4.1. Setup

As shown in Figure 3. The 123-node test system consists of two feeders, with substations located at nodes 150 and 451. It includes two DGs connected at nodes 195 and 350, each with a rated capacity of 1.5 MVA. The nodal voltage magnitudes are constrained within 0.95–1.05 p.u. The system also includes three DCs located at nodes 123, 124, and 251, each with a load demand of 1 MW. In addition, 18 RCSs are installed, and the line capacity limit is set to 5 MVA. Under normal operating conditions, the switches on branches 13–152, 151–300, and 54–94 are open, while all other switches remain closed. τ^{FI} and τ^{SR} are set to 0.5 h and 5 h, respectively.

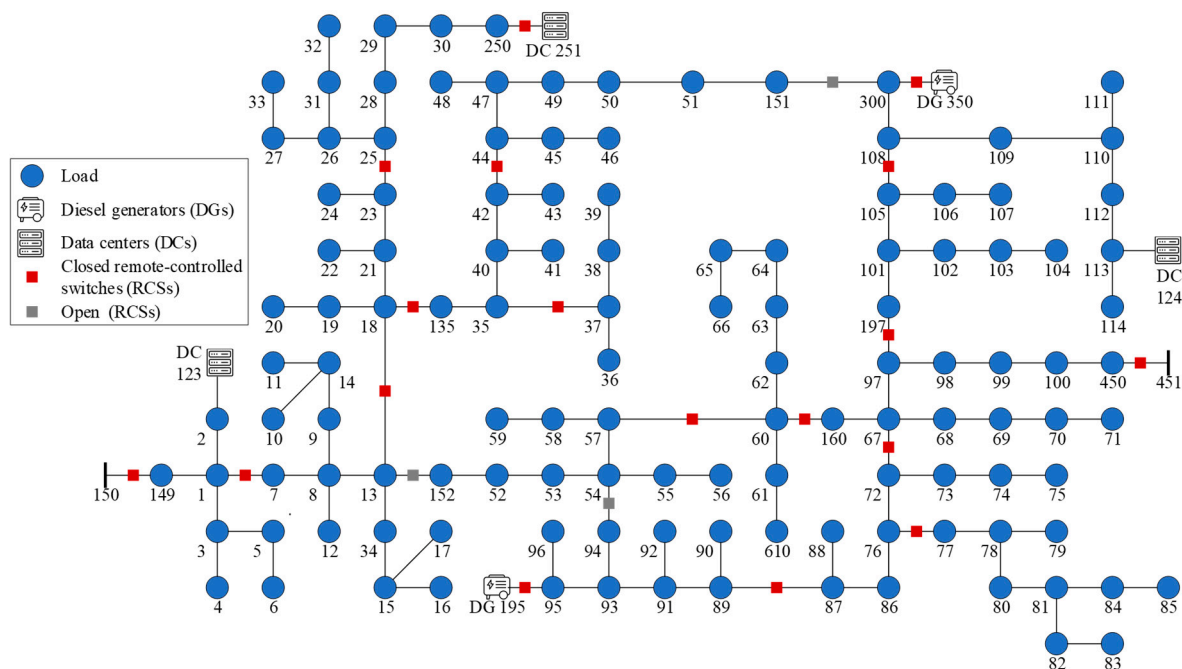


Figure 3. The modified IEEE 123-node test system.

The modified 356-node distribution system consists of three feeders, 356 nodes, and 357 branches. Among these branches, 35 are equipped with RCSs. The system includes five DGs, each with a rated capacity of 2.5 MVA. In addition, five DCs are integrated into the system, each with a load demand of 1.5 MW.

For the DC model, the indoor temperature is constrained within 5–25 °C to ensure desirable hardware performance and energy efficiency. The maximum allowable workload delay is set to 500 ms. The ratio of delay-tolerant workloads to latency-sensitive workloads is set to 4:1. This setting ensures real-time service requirements while prioritizing delay-tolerant workloads to reduce energy costs.

4.2. Modified IEEE-123 Node Test System

(1) Ablation analysis of network reconfiguration and DC workload migration

To more explicitly distinguish the contribution of network reconfiguration from that of DC workload migration, an ablation analysis is conducted in this subsection. Two cases are compared under the same fault scenarios, system topology, and operational constraints. In Case 1, only network reconfiguration is considered. Specifically, fault isolation and service restoration are optimized through switching operations, while DC workload migration is disabled. Therefore, each DC must be supplied locally after faults. In Case 2, the full proposed model is considered, in which network reconfiguration and DC workload migration are jointly optimized. Since network reconfiguration is included in both cases, the difference between Case 1 and Case 2 directly quantifies the additional reliability benefit provided by DC workload flexibility.

As shown in Table 1, SAIFI, SAIDI, and ASAI remain unchanged in the two cases, whereas EENS decreases from 3.8619 to 1.3895 when DC workload migration is enabled. This corresponds to a reduction of approximately 64.0%. Since both cases include the same network reconfiguration model, this reduction reflects the marginal contribution of DC workload migration under optimized post-fault switching operations.

Table 1. Ablation comparison between network reconfiguration only and the full model with DC workload migration.

Index	Case 1: Network Reconfiguration Only	Case 2: Network Reconfiguration + DC Workload Migration
SAIFI	0.0715	0.0715
SAIDI	0.2024	0.2024
ASAI	99.9976%	99.9976%
EENS	3.8619	1.3895

The physical interpretation of this result is related to the different meanings of the reliability indices. SAIFI and SAIDI mainly characterize the interruption frequency and outage duration of physical load points. They are determined by the fault scenarios, the fault isolation process, the service restoration topology, and the energization status of nodes. In the present case, enabling DC workload migration does not change the fault isolation logic, the switching sequence, or the final restored topology of the physical distribution network. Therefore, the set of interrupted physical load points and their outage durations remains unchanged. As a result, SAIFI and SAIDI are almost unaffected. Since ASAI is directly related to the accumulated outage duration, it also remains unchanged.

By contrast, EENS measures the amount of unsupplied energy. It is therefore sensitive not only to whether a node is energized, but also to the magnitude of the load that remains effectively unsupplied. DC workload migration provides cyber-layer flexibility by transferring part of the computational workload from an affected DC to other healthy DCs with available power supply and computing capacity. In this way, even if the original DC node is still electrically interrupted, part of its computational service can be maintained at remote DCs. Therefore, the equivalent unsupplied DC demand is reduced, which directly decreases EENS. This explains why DC workload migration significantly reduces EENS while having little effect on SAIFI and SAIDI in the present case.

It should be noted that this observation is case-dependent and does not imply that DC workload migration can never affect interruption frequency or outage duration indices. In more strongly coupled operating conditions, workload migration may also influence SAIFI or SAIDI. For example, if DC service continuity is explicitly counted as part of customer interruption statistics, successful workload migration may reduce the interruption frequency

or duration experienced by DC-related services. In addition, if migrating DC workload releases electrical capacity in a constrained feeder or islanded area, it may enable the restoration of additional conventional loads or shorten the duration of load interruption. Furthermore, when communication delay, bandwidth degradation, or staged restoration time is explicitly modeled, the feasibility and timing of workload migration may affect the service restoration sequence and thus influence duration-related reliability indices. These cases will be investigated in future work with more detailed cyber-physical restoration modeling.

To further illustrate the separate roles of network reconfiguration and DC workload migration, a representative fault scenario is analyzed. In this scenario, branch 23–21 is faulted. As shown in Figure 4, during the fault isolation stage, the switches on branches 13–18, 18–135, and 23–15 are opened to isolate the faulted section. Several downstream loads are disconnected from the healthy part of the network.

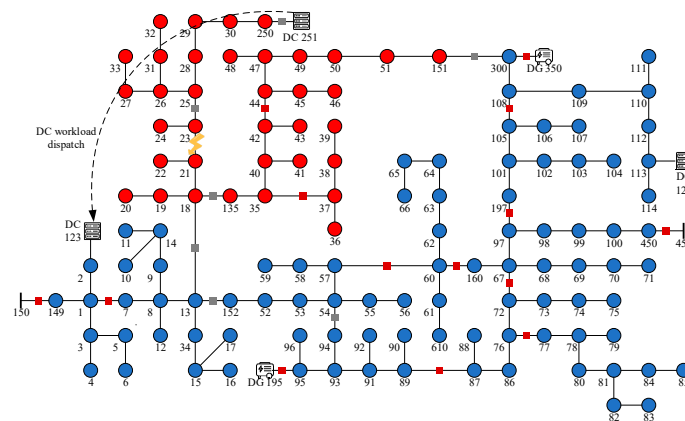


Figure 4. Fault isolation results for the studied scenario with a fault on branch 23–21. Red nodes indicate de-energized nodes, whereas blue nodes indicate energized nodes.

After fault isolation, service restoration is carried out through network reconfiguration. As shown in Figure 5, the restored topology shows that the switch on branch 151–300 is closed to transfer part of the interrupted loads to an alternative supply path. This confirms the role of network reconfiguration in reducing the outage area. However, due to network topology and limited restoration resources, network reconfiguration alone cannot fully restore all interrupted loads.

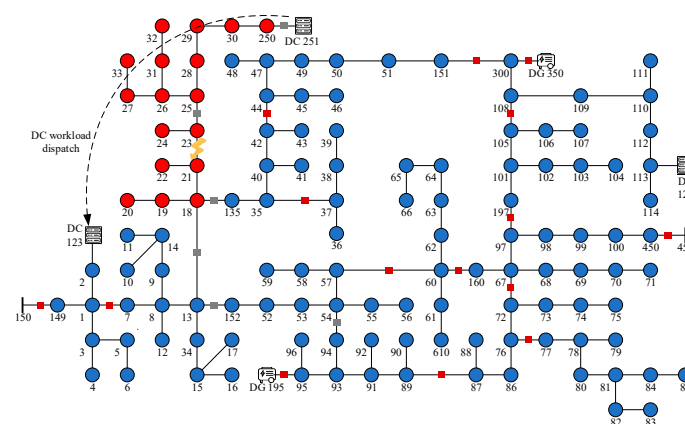


Figure 5. Restored network topology after service restoration for the studied scenario.

The quantitative results of the representative fault scenario are summarized in Table 2. As shown in Table 2, when only network reconfiguration is considered, the remaining interrupted load is 1.36 MW. When DC workload migration is further enabled, the interrupted

load is reduced to 0.36 MW. Therefore, DC workload migration provides an additional reduction of 1.00 MW, corresponding to approximately 73.5%. This result shows that network reconfiguration and DC workload migration contribute to reliability improvement through different mechanisms. Network reconfiguration restores loads by modifying the physical power supply path, whereas DC workload migration reduces the effective unsupplied demand by reallocating computational tasks among interconnected DCs.

Table 2. Marginal benefit of DC workload migration under network reconfiguration.

	Case 1: Network Reconfiguration Only	Case 2: Network Reconfiguration + DC Workload Migration
Interrupted load	1.36 MW	0.36 MW

The detailed workload transfer result is reported in Table 3. As shown in Table 3, 1 MW of workload originally assigned to DC 251 is transferred to DC 123. This indicates that DC 123 has sufficient remaining power supply and service capacity to accommodate the migrated workload after the fault. Consequently, the interruption of DC 251 does not directly lead to the same amount of unsupplied energy. This result verifies that geographically distributed DCs can provide additional spatial flexibility beyond physical network reconfiguration.

Table 3. Workload transfer paths of DCs in Case 2.

Workload Transfer Path	Transferred Load
DC 251 → DC 123	1 MW

Overall, this ablation analysis explicitly separates the contribution of network reconfiguration from that of DC workload migration. The former improves service restoration by changing the electrical supply path, while the latter further reduces the effective unsupplied DC demand through computational workload redistribution. This confirms that incorporating DC workload migration into the reliability assessment framework is necessary for accurately quantifying the resilience contribution of flexible DC loads.

(2) Sensitivity analysis of multiple heterogeneous faults

To quantify the impact of multiple heterogeneous faults, a sensitivity analysis is conducted in this subsection. Four cases are compared under the same distribution network, DC configuration, and restoration model. Case 2 represents the full proposed model, where line faults, node faults, and RCS failures are all considered. Case 3 considers only conventional line faults. Case 4 considers line faults and node faults but excludes RCS failures. Case 5 considers line faults and RCS failures but excludes node faults. Therefore, the comparison among these four cases identifies both the overall reliability impact of heterogeneous faults beyond conventional line outages and the marginal contribution of node faults and RCS failures.

As shown in Table 4, the reliability assessment results become more severe when node faults and RCS failures are included. Compared with Case 3, Case 2 increases SAIFI from 0.0473 to 0.0715, SAIDI from 0.0558 to 0.2024, and EENS from 0.3589 to 1.3895. The corresponding increases are approximately 51.2%, 262.7%, and 287.2%, respectively. Meanwhile, ASAI decreases from 99.9999% to 99.9976%. These results confirm that reliability assessment considering only conventional line faults may produce overly optimistic results.

Table 4. Sensitivity analysis of reliability indices under different fault types.

Index	Case 2: Multiple Heterogeneous Faults	Case 3: Line Faults Only	Case 4: Line Faults + Node Faults	Case 5: Line Faults + RCS Failures
SAIFI	0.0715	0.0473	0.0660	0.0528
SAIDI	0.2024	0.0558	0.1705	0.0877
ASAI	99.9976%	99.9999%	99.9981%	99.9990%
EENS	1.3895	0.3589	1.0527	0.6957

The fault-type decomposition further shows that node faults have the greatest marginal impact on the severity-oriented indices, especially SAIDI and EENS. When node faults are added to the line-fault-only case, EENS increases from 0.3589 to 1.0527, corresponding to an increase of approximately 193.3%. SAIDI increases from 0.0558 to 0.1705, corresponding to an increase of approximately 205.6%. By contrast, when RCS failures are added to the line-fault-only case, EENS increases from 0.3589 to 0.6957, and SAIDI increases from 0.0558 to 0.0877. Therefore, in the studied system, node faults impose a larger impact on outage duration and unsupplied energy than RCS failures.

This result is physically intuitive. A line fault usually affects the downstream area associated with the failed branch and can often be isolated by adjacent switches. In contrast, a node fault directly de-energizes the failed node and may also affect all connected incident branches. If the failed node is connected to a load, a DC, or a DG, the fault may simultaneously reduce load supply, workload migration capability, or local generation support. Therefore, node faults tend to enlarge the affected area and increase the amount of unsupplied energy.

RCS failures also have a non-negligible impact, although their effect is mainly reflected in the restriction of restoration flexibility. An RCS failure may prevent the intended switching operation during fault isolation or service restoration. Consequently, the available reconfiguration paths may be reduced, and some non-faulted loads may remain disconnected. This explains why Case 2 produces higher SAIDI and EENS than Case 4. In particular, when RCS failures are further considered on the basis of line and node faults, EENS increases from 1.0527 to 1.3895, corresponding to an increase of approximately 32.0%.

Table 5 summarizes the incremental impact of different fault mechanisms. Line faults provide the dominant baseline contribution to interruption frequency because distribution feeders contain many branches and line outages are the most common contingency type in conventional reliability assessment. However, among the additional heterogeneous fault types, node faults have the greatest impact on SAIDI and EENS. This is because node faults directly interrupt the associated load point and may also affect connected DCs, DGs, and adjacent branches. RCS failures mainly affect the restoration process by limiting available switching actions. Therefore, their impact is smaller than that of node faults in this case, but it remains significant for restoration-oriented reliability evaluation.

Table 5. Incremental impact of different fault types.

Fault Mechanism	Comparison Basis	Δ SAIFI	Δ SAIDI	Δ EENS	Main Impact
Line faults	Case 3	0.0473	0.0558	0.3589	Primary source of interruption frequency
Node faults	Case 4–Case 3	0.0187	0.1147	0.6938	Largest impact on outage duration and unsupplied energy
RCS failures	Case 2–Case 4	0.0055	0.0319	0.3368	Reduced switching and restoration flexibility

Overall, the results indicate that different fault types affect reliability indices through different mechanisms. Line faults mainly determine the baseline interruption frequency. Node faults significantly increase outage duration and unsupplied energy because they directly affect nodal supply and connected resources. RCS failures reduce the effectiveness of post-fault reconfiguration and thereby worsen restoration performance. These findings confirm the necessity of explicitly modeling multiple heterogeneous faults in reliability assessment. The proposed framework provides a more realistic evaluation by incorporating these fault mechanisms into the optimization-based restoration model.

(3) Comparison with a rule-based heuristic restoration strategy

To further demonstrate the advantage of the proposed optimization-based reliability assessment framework, a representative rule-based heuristic restoration strategy is introduced as a benchmark. The heuristic method follows a sequential restoration logic commonly used in conventional reliability assessment. After fault isolation, candidate switching actions are ranked according to load priority and electrical distance to the nearest available source. In this study, the load priority is assigned according to the amount of restorable active load, since no differentiated customer interruption cost is specified in the test system. The electrical distance is measured by the shortest available path length from the candidate restorable section to an energized source, where faulted components and unavailable switches are excluded. If several candidate switching actions have the same ranking, the action that restores a larger amount of load is selected first; if a tie still exists, the action with the smaller branch index is selected to ensure deterministic implementation.

The tie switches are then closed sequentially if the resulting topology satisfies radiality, voltage, branch capacity, and DG output constraints. After each accepted switching action, DG outputs are adjusted within their capacity limits to supply the restored loads. If the available supply or network capacity is insufficient, the residual load shedding is determined according to the same priority order, with lower-priority loads shed first. Once the network topology is determined, the remaining interrupted DC workload is transferred greedily to healthy DCs with available power supply and service capacity, subject to the workload transfer constraints. When multiple healthy DCs are available, the receiving DC with the largest remaining service capacity is selected first; if a tie exists, the DC with the shorter transfer path is selected. Different from the proposed method, this heuristic strategy does not jointly optimize network reconfiguration and DC workload migration.

The comparison results are summarized in Table 6. The same fault scenarios, component failure assumptions, DC parameters, and operational constraints are used for both methods.

Table 6. Comparison between the proposed optimization-based method and the rule-based heuristic restoration method.

Index	Rule-Based Heuristic Restoration	Proposed Optimization-Based Method
SAIFI	0.0715	0.0715
SAIDI	0.2598	0.2024
ASAI	99.9970%	99.9976%
EENS	2.1276	1.3895

As shown in Table 6, the proposed optimization-based method achieves better reliability performance than the rule-based heuristic method. The SAIFI values of the two methods are identical because both methods consider the same fault scenarios and fault isolation process. Therefore, the number of customers initially interrupted by fault events remains unchanged. However, the proposed method achieves lower SAIDI and EENS. Specifically, SAIDI decreases from 0.2598 to 0.2024, corresponding to a reduction of ap-

proximately 22.1%. EENS decreases from 2.1276 to 1.3895, corresponding to a reduction of approximately 34.7%.

The improvement is mainly attributed to the joint optimization of post-fault switching operations and DC workload migration. The heuristic strategy restores loads according to a predefined priority and local feasibility checks. Although this strategy can generate feasible restoration decisions, it may fail to identify the globally preferable combination of switching actions and workload transfer paths. In contrast, the proposed MILP-based framework optimizes all restoration decisions simultaneously under network security, radiality, DG operation, and DC workload transfer constraints. As a result, it can more effectively utilize available network capacity and DC spatial flexibility.

These results confirm that the proposed optimization-based framework provides a more effective post-fault restoration assessment than conventional heuristic restoration logic. The comparison also supports the necessity of formulating reliability assessment as an explicit optimization problem, especially for active distribution systems with flexible DC loads and heterogeneous fault conditions.

4.3. Modified 356-Node Regional System in China

To further evaluate the scalability of the proposed MILP-based reliability assessment framework, an additional case study is conducted on a modified 356-node distribution system. In this test, the same modeling framework, restoration logic, and reliability evaluation procedure as those in the modified IEEE 123-node system are adopted. Multiple heterogeneous faults are considered, including line faults, node faults, and RCS failures. The reliability assessment results are summarized in Table 7.

Table 7. Reliability assessment results of the modified 356-node distribution system.

	SAIFI	SAIDI	ASAI	EENS
Modified 356-node system	0.1168	0.3427	99.9961%	5.2846

As shown in Table 7, the proposed method remains applicable to the larger 356-node system. Compared with the modified IEEE 123-node system, the reliability indices become worse because the larger system contains more network components, more fault scenarios, and a larger amount of DC demand. Specifically, the increased number of branches and nodes increases the exposure to component failures. Meanwhile, although additional RCSs and DGs improve restoration flexibility, the larger network scale also expands the post-fault search space for switching operations and workload migration. As a result, the evaluated SAIFI, SAIDI, and EENS are higher than those of the smaller test system, while ASAI decreases slightly. This trend is consistent with the expected reliability behavior of larger active distribution systems.

As shown in Table 8, the computational burden increases with the scale of the distribution system. Compared with the modified IEEE 123-node system, the modified 356-node system has approximately 2.9 times more nodes and branches, and the number of RCSs increases from 18 to 35. Accordingly, the average solution time per scenario increases from 1.24 s to 5.61 s. This increase is reasonable because a larger network introduces more power flow variables, nodal energization variables, branch status variables, and switching-related binary variables. In addition, the increased number of RCSs enlarges the feasible reconfiguration space, which increases the complexity of the MILP solution process.

Table 8. Computational comparison between the modified IEEE 123-node and 356-node systems.

	Nodes	Branches	RCSs	Average Solution Time per Scenario
Modified IEEE 123-node system	123	124	18	1.24 s
Modified 356-node system	356	357	35	5.61 s

Although the solution time increases with system size, the proposed method remains computationally tractable for the modified 356-node system. The average solution time remains within several seconds per scenario, indicating that the proposed MILP formulation applies to medium-scale distribution systems. The tractability is supported by three modeling choices. First, the nonlinear distribution power-flow equations are approximated using a linearized DistFlow formulation. Second, the radiality constraints are modeled using continuous auxiliary variables, which avoids introducing excessive additional binary variables. Third, the overall reliability assessment is separable across fault scenarios. Once a fault scenario is specified, the corresponding post-fault restoration problem can be solved independently. This structure enables scenario-wise parallel computation and can further accelerate the overall assessment. For very large-scale systems with extensive fault scenario enumeration, decomposition, scenario screening, and parallel implementation may be required.

5. Conclusions

This paper proposes an optimization model-based reliability assessment framework for active distribution systems with integrated data centers (DCs). The proposed framework co-optimizes network reconfiguration and DC workload migration during fault isolation and service restoration. It also incorporates multiple heterogeneous faults, including line faults, node faults, and remote-controlled switch failures. Based on a detailed DC workload dispatch model, the reliability assessment problem is reformulated as a mixed-integer linear programming problem.

Case studies on a modified IEEE 123-node test feeder demonstrate the effectiveness of the proposed method. The results show that DC workload migration reduces EENS from 3.8619 to 1.3895, corresponding to a reduction of approximately 64.0%. This indicates that interconnected DCs can reduce unsupplied energy by reallocating computational workloads among geographically distributed sites. The results also show that neglecting node faults and RCS failures leads to overly optimistic reliability estimates. An additional test on a modified 356-node regional distribution system further verifies the applicability of the proposed framework to a larger system.

Overall, the proposed method provides a physically interpretable reliability assessment tool for distribution systems with DC integration and heterogeneous fault mechanisms. Future work will further consider time-varying DC workloads, differentiated load criticality coefficients, more detailed RCS failure modes, communication delays and failures, and decomposition or parallel optimization techniques for very large-scale systems.

Future work will focus on several directions. First, the current model can be extended to incorporate time-varying DC workloads and more detailed workload characteristics, such as task priorities, latency constraints, and temporal migration flexibility. In addition, differentiated load criticality coefficients may be introduced for conventional loads, critical loads, and DC computational workloads. This extension would allow the model to explicitly represent the priority trade-off between restoring local electrical demand and migrating computational tasks under constrained restoration resources and limited communication bandwidth. Second, more diverse, flexible resources, including energy storage systems, demand response, and renewable generation uncertainty, can be integrated into the reliability assessment framework. Third, the modeling of RCS failures can be further refined. In

the present study, an RCS failure is represented as the inability of the switch to change its normal operating status. Future work may consider more realistic failure modes, including stuck-open failures, stuck-closed failures, actuator failures, delayed switching actions, and communication-induced operation failures. This extension would allow the reliability impact of automation-device failures to be evaluated more comprehensively. Fourth, more detailed information and communication system models should be incorporated to better characterize cyber-physical coupling in modern distribution systems. In particular, communication delay, communication-link outage, command transmission failure, bandwidth degradation, and delayed or failed DC workload migration may be explicitly modeled. Finally, to improve scalability for larger practical systems, future studies may investigate decomposition, acceleration, parallel computing, scenario screening, or learning-assisted optimization methods for reliability assessment under a large number of fault scenarios.

Author Contributions: W.Z.: Conceptualization, Methodology, Software, Writing—original draft, Visualization. Y.L.: Methodology, Software, Validation, Formal analysis. W.H.: Investigation, Resources, Data curation. L.H.: Formal analysis, Validation, Writing—review and editing. K.P.: Software, Data curation, Visualization. S.L.: Investigation, Resources, Validation. Q.L.: Resources, Project administration, Writing—review and editing. T.Y.: Supervision, Project administration, Funding acquisition, Writing—review and editing. W.X.: Conceptualization, Methodology, Supervision, Funding acquisition, Writing—review & editing. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the Special Research Project on Guangdong Power Grid Planning, grant number [030200QQ00250002].

Data Availability Statement: The original contributions presented in this study are included in the article. Further inquiries can be directed to the corresponding author.

Conflicts of Interest: Authors Weiliang Zhong, Yongbiao Liang, Wei Huang, Lieliang Hu, Kaining Pan, and Sineng Li were employed by the company Shaoguan Power Supply Bureau of Guangdong Power Grid Co., Ltd. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Abbreviations

Abbreviations

DCs	Data centers.
DGs	Distributed generators.
RCSs	Remote-controlled switches.
MILP	Mixed-integer linear programming.
NO	Normal operation.
FI	Fault isolation.
SR	Service restoration.
SAIDI	System average interruption duration index.
SAIFI	System average interruption frequency index.
ASAI	Average service availability index.
EENS	Expected energy not supplied.

Indices

s	Index for fault scenarios.
i, j, k	Index for nodes in the distribution system.
ij, ji	Index for branches in the distribution system.

Sets

S	Set of all fault scenarios.
F	Set of all faults, including branches, nodes, and RCSs.
Ω_N	Set of all nodes in the distribution system.
Ω_S	Set of all substation nodes in the distribution system.

PH	Sets of phases including FI and SR.
Parameters	
λ^{DC}	Power usage effectiveness coefficient of the DC.
$P_i^{DC,max}$	Maximum power capacity limit of the DC at node i .
$W_{i,s,t}$	Computational workload assigned to the DC.
n_i^{IT}	Total number of installed servers of the DC at node i .
ϕ	Backup coefficient of DCs.
S_i^{max}	Upper bound of the CPU utilization for a single server.
φ_i	Processing capacity of each server unit.
$H_{i,s,t}^{cool}$	Heat load generated by the IT equipment at node i in scenario s at time t
λ^{cool}	Coefficient of performance of the cooling system.
$P_i^{cool,max}$	Rated power capacity of the installed cooling infrastructure at node i .
B_{ij}	Available communication bandwidth between DC i and DC j .
p_s	Occurrence probability of scenario s .
$F_{ij,s}^B$	Fault states of branch ij in scenario s . 1 indicates that the branch is faulty, 0 otherwise.
$F_{i,s}^N$	Fault states of node i in scenario s . 1 indicates that the node is faulty, 0 otherwise.
$F_{ij,s}^{RCS}$	Fault states of RCS on branch ij in scenario s . 1 indicates that the RCS is faulty, 0 otherwise.
r_{ij}/x_{ij}	Resistance and inductance of branch ij .
M	A large number.
$P_i^{DG,max}$	Maximum active power outputs of the DG at node i .
$P_i^{DG,min}$	Minimum active power outputs of the DG at node i .
$Q_i^{DG,max}$	Maximum reactive power outputs of the DG at node i .
$Q_i^{DG,min}$	Minimum reactive power outputs of the DG at node i .
S_{ij}^{max}	Maximum capacity of branch ij .
U_i^{max}/U_i^{min}	Upper and lower bounds of the voltage at node i .
x_{ij}^{RCS}	Binary: 1 if an RCS is installed on branch ij , 0 otherwise.
z_{ij}^{NO}	Status of branch ij at the normal operation phase.
B_{ij}^{DC}	Maximum transferable equivalent DC load from DC i to DC j , converted from communication bandwidth and workload-power relationship.
Variables	
$P_{i,s,t}^{DC}$	Total power demand of the DC at node i in scenario s at time step t .
$P_{i,s,t}^{IT}$	Server power consumption of the DC at node i in scenario s at time step t .
$P_{i,s,t}^{cool}$	Cooling power of the DC at node i in scenario s at time step t .
P_i^{IT-i}	Idle power consumption of a single server.
P_i^{IT-p}	Peak power consumption of a single server.
$n_{i,s,t}^{IT-on}$	Number of active servers of the DC at node i in scenario s at time step t .
$P_{ij,s,t}^{DC}$	Workload transferred from DC i to DC j .
ENS_s^{PH}	Unsupplied energy in scenario s at stage PH .
$P_{i,s,t}^{L,PH}$	Load demand at node i in scenario s at stage PH .
$y_{i,s}^{PH}$	Binary: 1 if node i is energized, 0 otherwise.
$P_{ik,s}^{PH}/Q_{ik,s}^{PH}$	Active and reactive power flow on branch ik during the PH phase of scenario s .
$P_{i,s}^{S,PH}/Q_{i,s}^{S,PH}$	Active and reactive power injection from the substation at node i during the PH phase of scenario s .
$P_{i,s}^{DG,PH}/Q_{i,s}^{DG,PH}$	Active and reactive power injection from the DG at node i during the PH phase of scenario s .
$P_{i,s}^{L,PH}/Q_{i,s}^{L,PH}$	Active and reactive power demand at node i during the PH phase of scenario s .
$P_{i,s}^{DC,PH}/Q_{i,s}^{DC,PH}$	Active and reactive power demand of the DC at node i during the PH phase of scenario s .

$U_{i,s}^{PH}$	Square of voltage magnitude at node i during the PH phase of scenario.
$z_{ij,s}^{PH}$	Binary: 1 if the RCS on branch ij is closed during the PH phase of scenarios s .
$f_{ij,s}^{+,PH} / f_{ij,s}^{-,PH}$	Auxiliary variables for the radial constraint on branch ij during the PH phase of scenario s .
$n_{i,s}^{FI}$	Binary: 1 if node i is involved in a faulted zone, 0 otherwise.

References

- Li, B.; Wang, S.; Chen, Y.; Han, Y.; Yang, W. Joint network expansion planning with internet data center and charging stations for EV sharing system. *IEEE Trans. Ind. Inform.* **2024**, *20*, 13251–13261. [\[CrossRef\]](#)
- Dong, H.; Wang, L.; Zhang, X.; Zeng, M. A two-stage stochastic collaborative planning approach for data centers and distribution network incorporating demand response and multivariate uncertainties. *J. Clean. Prod.* **2024**, *451*, 141482. [\[CrossRef\]](#)
- Wan, J.; Zhou, J.; Gui, X. Sustainability analysis of green data centers with CCHP and waste heat reuse systems. *IEEE Trans. Sustain. Comput.* **2021**, *6*, 155–167. [\[CrossRef\]](#)
- Huang, Y.; Deng, Z.; Chen, Y.; Zhang, C. Performance investigation of a biomimetic latent heat thermal energy storage device for waste heat recovery in data centers. *Appl. Energy* **2023**, *335*, 120745. [\[CrossRef\]](#)
- Song, M.; Yan, X.Y.; Wang, X.D.; Gao, C.; Zhang, W.; Xia, S. Optimal scheduling strategy of networked data centers in multiple transactive energy markets. *CSEE J. Power Energy Syst.* **2024**, early access. [\[CrossRef\]](#)
- Chen, M.; Gao, C.W.; Song, M.; Chen, S.; Li, D.; Liu, Q. Internet data centers participating in demand response: A comprehensive review. *Renew. Sustain. Energy Rev.* **2020**, *117*, 109466. [\[CrossRef\]](#)
- Wang, Z.; Wang, Y.; Ji, H.; Hasanien, H.M.; Zhao, J.; Yu, L.; He, J.; Yu, H.; Li, P. Distributionally robust planning for data center park considering operational economy and reliability. *Energy* **2024**, *290*, 130185. [\[CrossRef\]](#)
- Chen, B.; Che, Y.; Takci, M.T.; Qadrdan, M.; Zhou, Y. Spatial flexibility provision from geographically dispersed data centers enabling coordinated operation of multi-local flexibility markets. *IEEE Trans. Smart Grid* **2026**, *17*, 1371–1381. [\[CrossRef\]](#)
- Jian, J.; Zhao, J.; Ji, H.; Bai, L.; Xu, J.; Li, P.; Wu, J.; Wang, C. Supply restoration of data centers in flexible distribution networks with spatial-temporal regulation. *IEEE Trans. Smart Grid* **2024**, *15*, 340–354. [\[CrossRef\]](#)
- Karngala, K.; Singh, C. Reliability assessment framework for the distribution system including distributed energy resources. *IEEE Trans. Sustain. Energy* **2021**, *12*, 1539–1548. [\[CrossRef\]](#)
- Zhang, D.; Li, G.; Bie, Z.; Fan, K. An analytical method for reliability evaluation of power distribution system with time-varying failure rates. *Reliab. Eng. Syst. Saf.* **2024**, *250*, 110290. [\[CrossRef\]](#)
- Li, G.; Zhang, D.; Bie, Z. An analytical method for reliability and resilience evaluation of power distribution systems with mobile power sources. *Reliab. Eng. Syst. Saf.* **2026**, *267*, 111817. [\[CrossRef\]](#)
- Weng, Z.; Zhou, J.; Zhan, Z. Reliability evaluation of standalone microgrid based on sequential Monte Carlo simulation method. *Energies* **2022**, *15*, 6706. [\[CrossRef\]](#)
- Huang, W.; Hu, B.; Shao, C.; Li, W.; Wang, X.; Xie, K.; Chung, C.Y. Power system reliability evaluation based on sequential Monte Carlo simulation considering multiple failure modes of components. *J. Mod. Power Syst. Clean Energy* **2025**, *13*, 202–214. [\[CrossRef\]](#)
- Karngala, K.; Singh, C.; Xie, L. Predictive reliability assessment of distribution grids with residential distributed energy resources. *CSEE J. Power Energy Syst.* **2025**, *11*, 2598–2609. [\[CrossRef\]](#)
- Dong, J.; Zhu, L.; Dong, Q.; Kritprajun, P.; Liu, Y.; Liu, Y.; Tolbert, L.M.; Hambrick, J.C.; Xue, Y.; Ollis, T.B.; et al. Integrating transactive energy into reliability evaluation for a self-healing distribution system with microgrid. *IEEE Trans. Sustain. Energy* **2022**, *13*, 122–134. [\[CrossRef\]](#)
- Muñoz-Delgado, G.; Contreras, J.; Arroyo, J.M. Reliability assessment for distribution optimization models: A non-simulation-based linear programming approach. *IEEE Trans. Smart Grid* **2018**, *9*, 3048–3059. [\[CrossRef\]](#)
- Li, Z.; Wu, W.; Tai, X.; Zhang, B. Optimization model-based reliability assessment for distribution networks considering detailed placement of circuit breakers and switches. *IEEE Trans. Power Syst.* **2020**, *35*, 3991–4004. [\[CrossRef\]](#)
- Jooshaki, M.; Abbaspour, A.; Fotuhi-Firuzabad, M.; Munoz-Delgado, G.; Contreras, J.; Lehtonen, M.; Arroyo, J.M. An enhanced MILP model for multistage reliability-constrained distribution network expansion planning. *IEEE Trans. Power Syst.* **2022**, *37*, 118–131. [\[CrossRef\]](#)
- Jooshaki, M.; Lehtonen, M.; Fotuhi-Firuzabad, M.; Muñoz-Delgado, G.; Contreras, J.; Arroyo, J.M. On the explicit formulation of reliability assessment of distribution systems with unknown network topology: Incorporation of DG, switching interruptions, and customer-interruption quantification. *Appl. Energy* **2022**, *324*, 119655. [\[CrossRef\]](#)
- Wang, Z.; Zhong, L.; Pan, Z.; Yu, T.; Qiu, X.; Wang, Y. Optimal double Q AC-DC hybrid distribution system planning with explicit topology-variable-based reliability assessment. *Appl. Energy* **2022**, *322*, 119438. [\[CrossRef\]](#)

22. Li, J.; Kong, H.; Wang, W.; Sun, L.; Xiao, W.; Wang, Z. A novel reliability assessment method for distribution networks based on linear programming considering distribution automation and distributed generation. *IET Renew. Power Gener.* **2024**, *18*, 529–544. [[CrossRef](#)]
23. Zhang, J.; Wang, B.; Ma, H.; He, Y.; Wang, Y.; Xue, Z. Reliability evaluation of cabled active distribution network considering multiple devices: A generalized MILP model. *Processes* **2023**, *11*, 3404. [[CrossRef](#)]
24. Wu, W.; Wang, C.; Wang, Y.; Dong, X.; Yang, M.; Zhang, Y.; Pouresmaeil, E.; Zhong, S.; Zhou, S.; Han, X. Reliability assessment of cyber physical distribution systems including sequential scenario simulation correction and recovery resource decisions. *Energy Convers. Econ.* **2026**, *7*, 37–49. [[CrossRef](#)]
25. Liu, J.; Qin, C.; Yu, Y. A comprehensive resilience-oriented FLISR method for distribution systems. *IEEE Trans. Smart Grid* **2021**, *12*, 2136–2152. [[CrossRef](#)]
26. Liu, J.; Qin, C.; Yu, Y. Enhancing distribution system resilience with proactive islanding and RCS-based fast fault isolation and service restoration. *IEEE Trans. Smart Grid* **2020**, *11*, 2381–2395. [[CrossRef](#)]
27. Wang, Z.; Lin, D.; Yu, T.; Zeng, G.; Pan, Z. Explicit reliability incorporated switches planning of smart distribution system with high flexibility. *Int. J. Power Energy Syst.* **2023**, *145*, 108629. [[CrossRef](#)]
28. Chen, X.; Wu, W.; Zhang, B. Robust restoration method for active distribution networks. *IEEE Trans. Power Syst.* **2016**, *31*, 4005–4015. [[CrossRef](#)]
29. Sun, S.; Li, G.; Chen, C.; Bian, Y.; Bie, Z. A novel formulation of radiality constraints for resilient reconfiguration of distribution systems. *IEEE Trans. Smart Grid* **2023**, *14*, 1337–1340. [[CrossRef](#)]
30. Zhao, T.; Wang, J. Learning Sequential Distribution System Restoration via Graph-Reinforcement Learning. *IEEE Trans. Power Syst.* **2022**, *37*, 1601–1611. [[CrossRef](#)]
31. Xiao, W.; Luo, Q.; Yu, T.; Wu, Y.; Huang, Z.; Pan, Z. Toward resilient distribution system via microgrid formation: A safe hierarchical hybrid reinforcement learning approach. *Appl. Energy* **2026**, *417*, 128001. [[CrossRef](#)]

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.